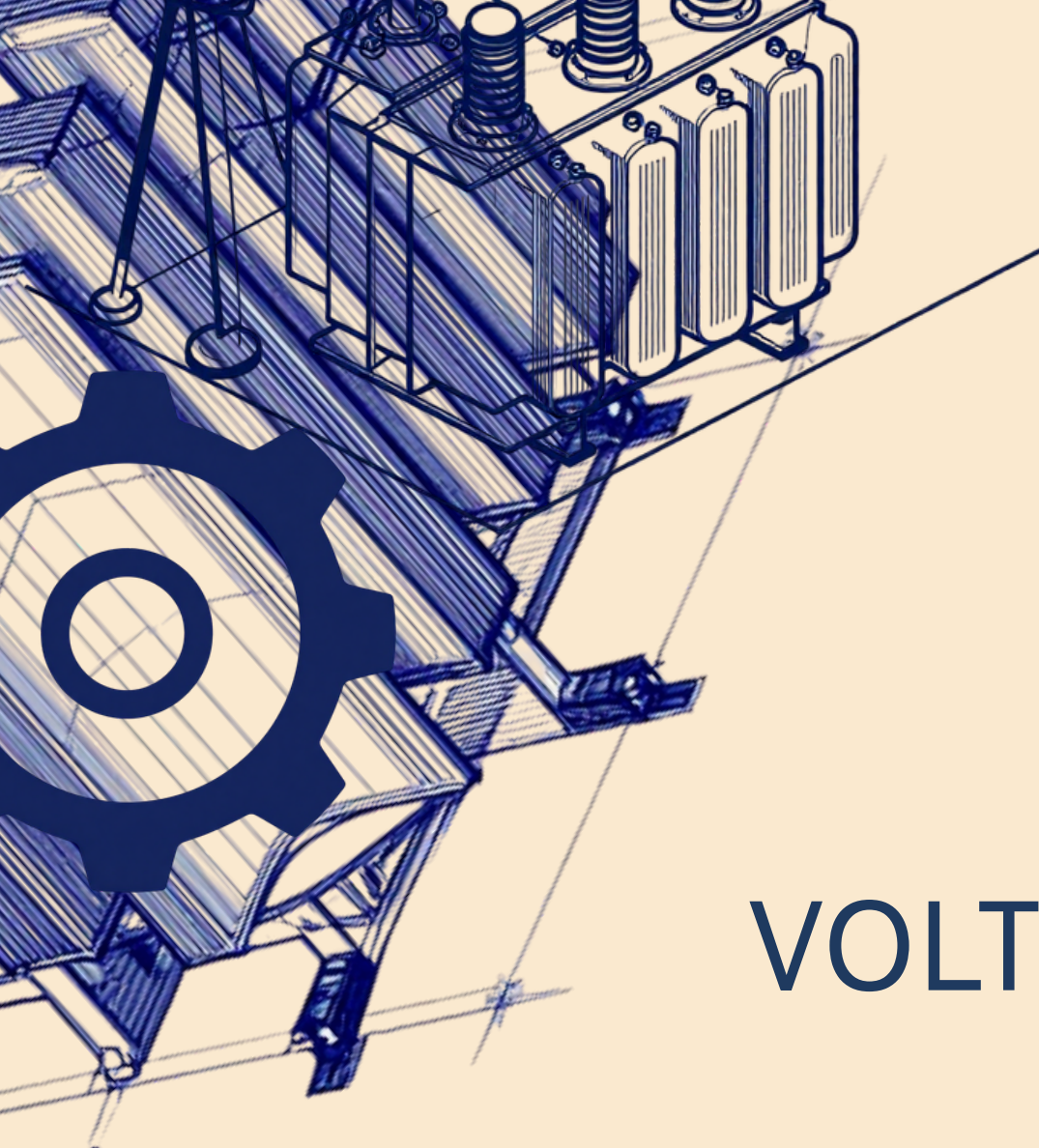




GRID RISK ADVISOR

VOLTRA: NEURAL GRID INTELLIGENCE ENGINE

PROBLEM



01 · DATA SILOS

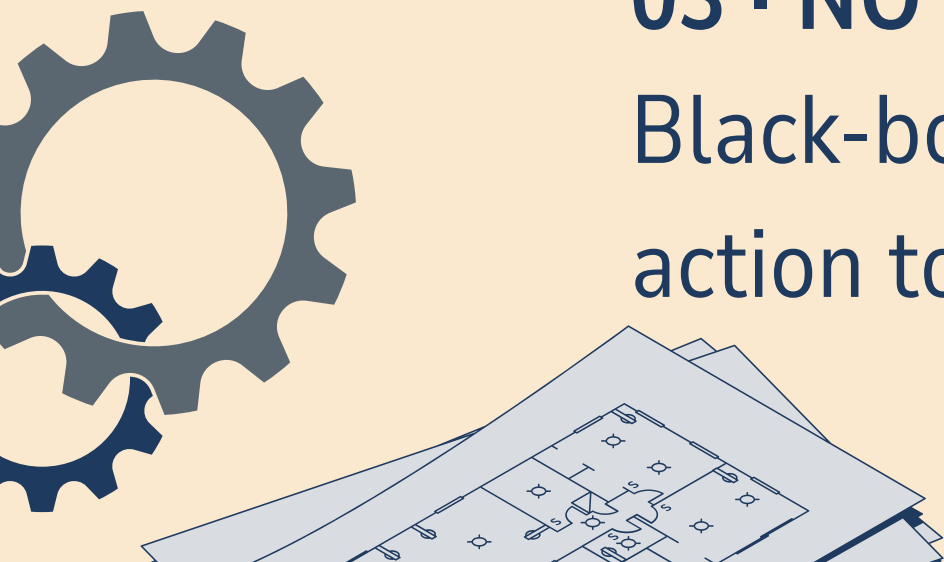
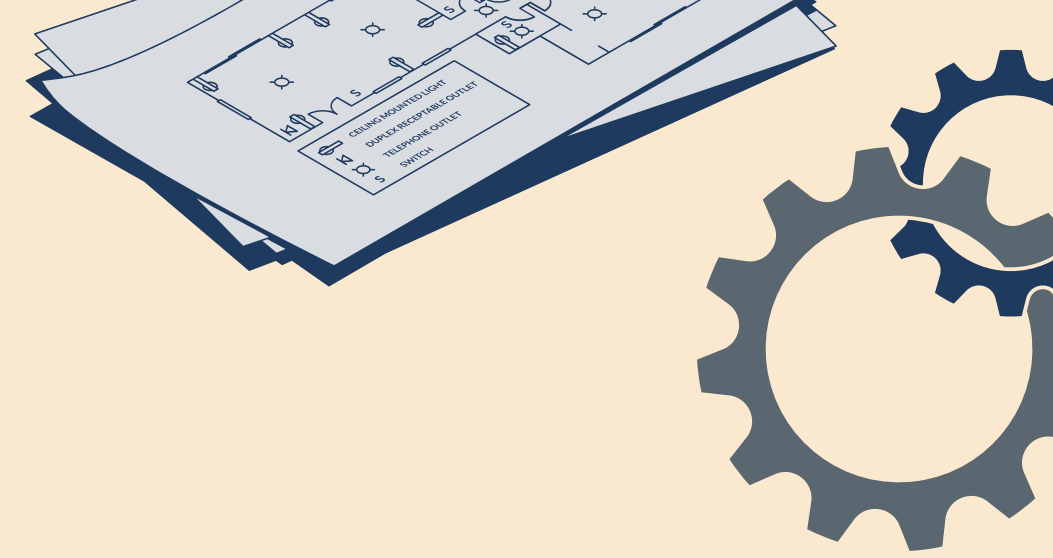
Critical data like DGA, temperature, vibration, and weather is scattered across systems.

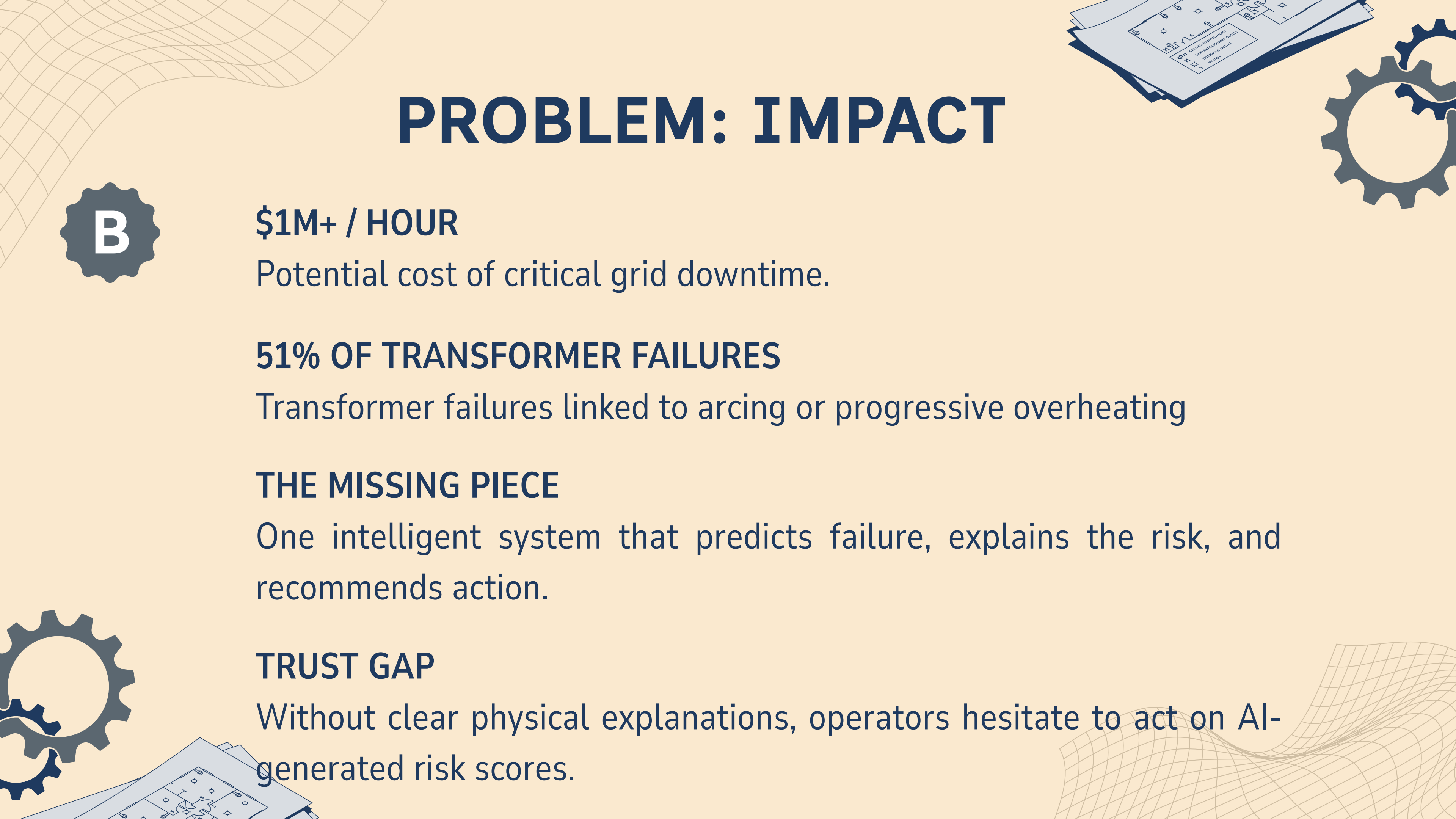
02 · REACTIVE MAINTENANCE

Utilities often respond after failure, making emergency intervention 3×–7× more expensive.

03 · NO CLEAR EXPLANATION

Black-box risk scores don't tell engineers what is failing, why, or what action to take.





PROBLEM: IMPACT

B

\$1M+ / HOUR

Potential cost of critical grid downtime.

51% OF TRANSFORMER FAILURES

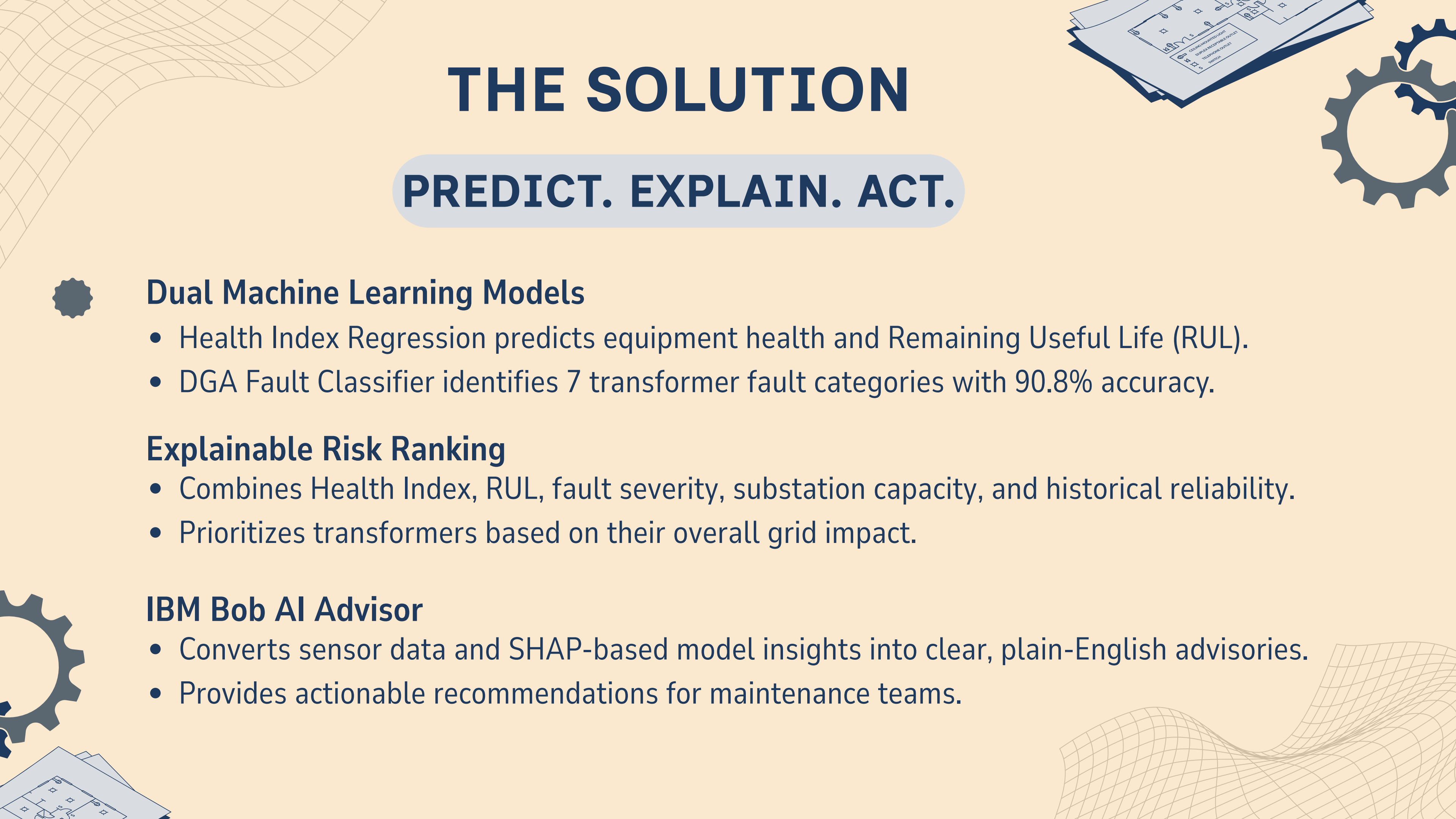
Transformer failures linked to arcing or progressive overheating

THE MISSING PIECE

One intelligent system that predicts failure, explains the risk, and recommends action.

TRUST GAP

Without clear physical explanations, operators hesitate to act on AI-generated risk scores.



THE SOLUTION

PREDICT. EXPLAIN. ACT.

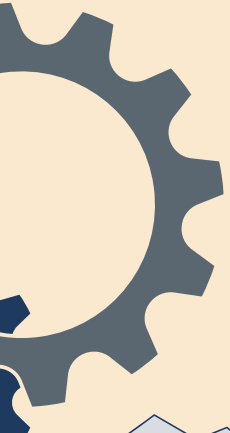


Dual Machine Learning Models

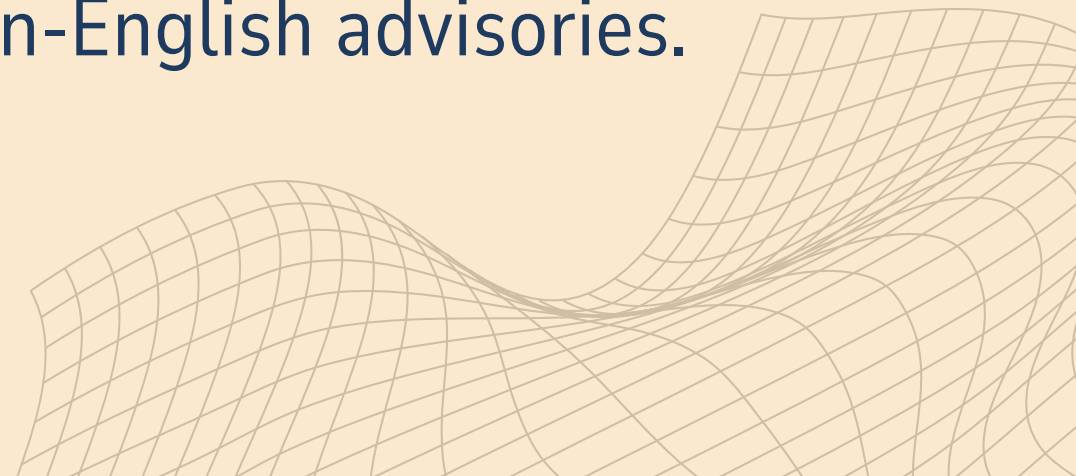
- Health Index Regression predicts equipment health and Remaining Useful Life (RUL).
- DGA Fault Classifier identifies 7 transformer fault categories with 90.8% accuracy.

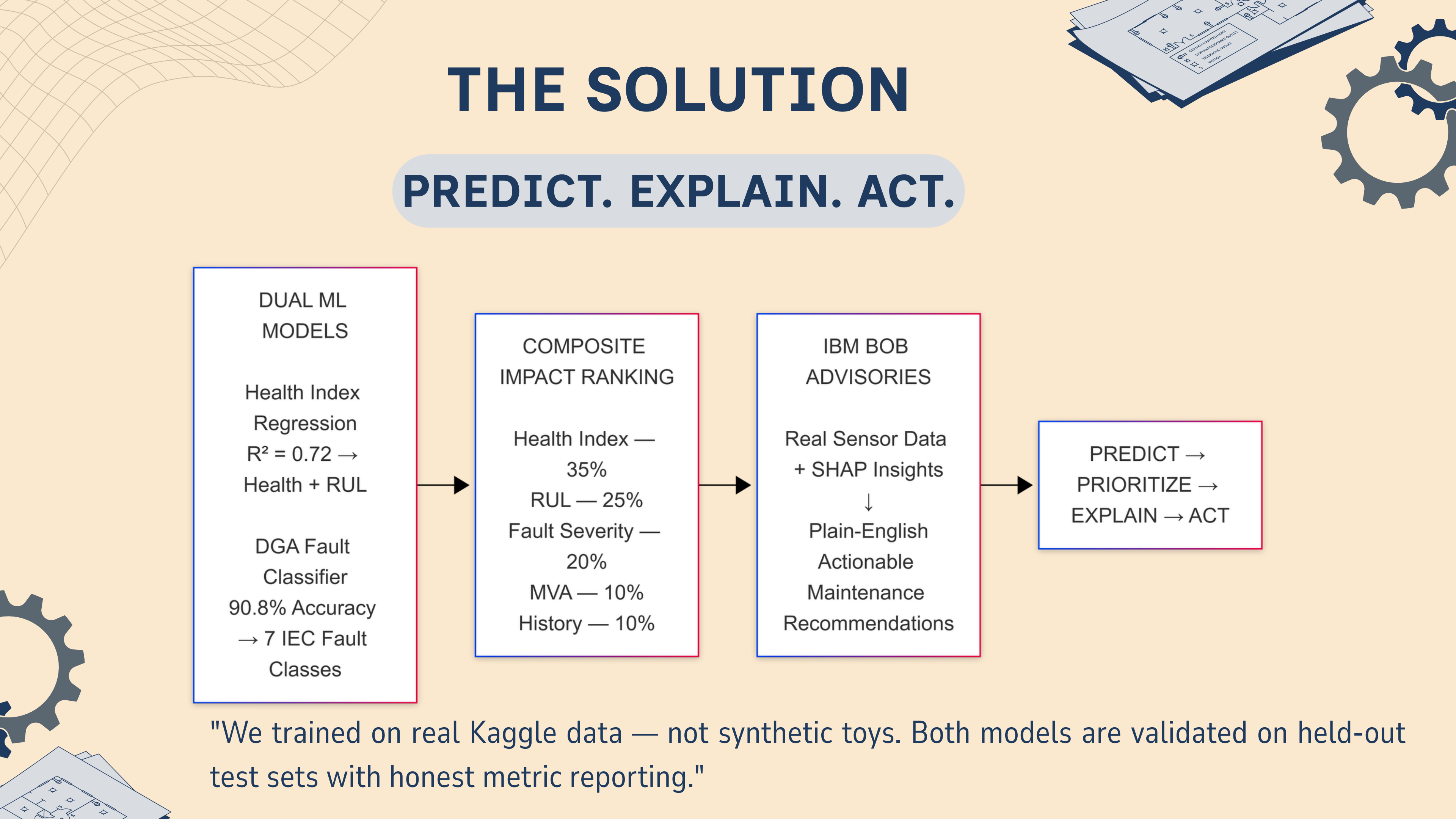
Explainable Risk Ranking

- Combines Health Index, RUL, fault severity, substation capacity, and historical reliability.
- Prioritizes transformers based on their overall grid impact.

A large, dark blue gear icon with eight teeth, positioned to the left of the 'IBM Bob AI Advisor' section header.

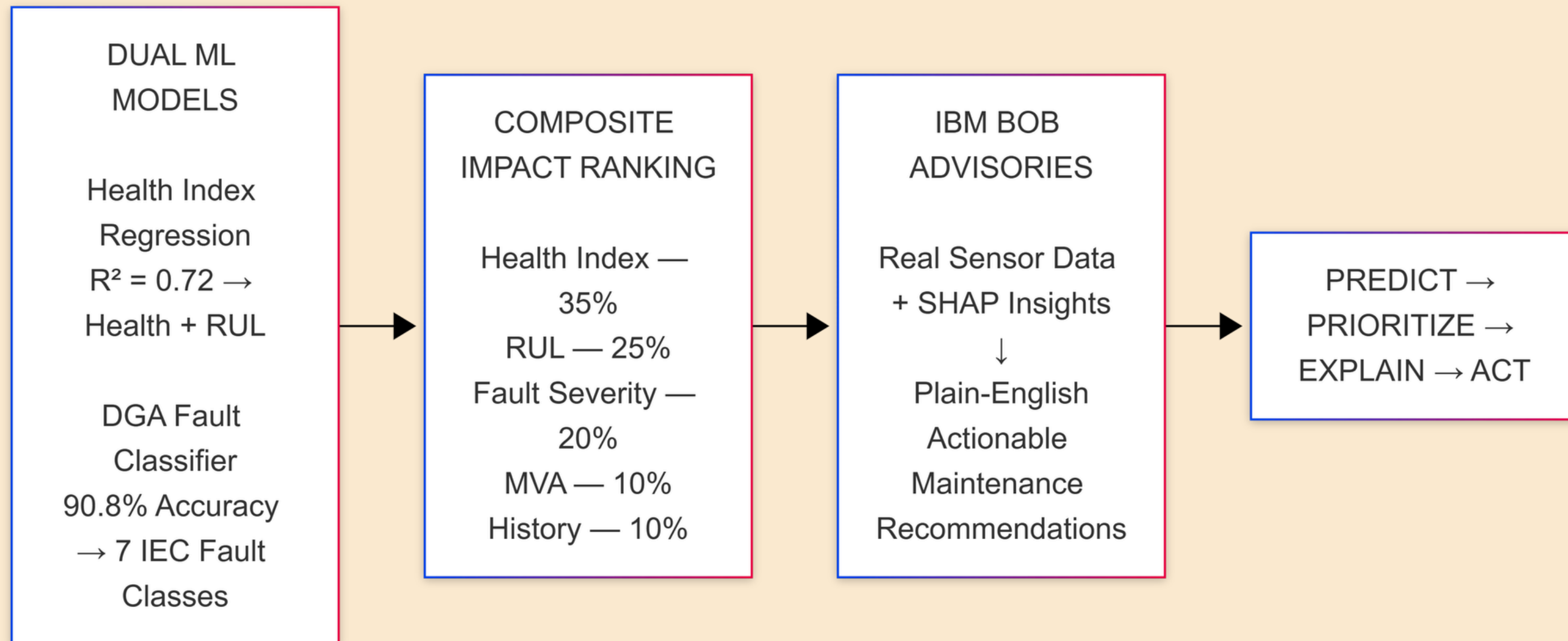
IBM Bob AI Advisor

- Converts sensor data and SHAP-based model insights into clear, plain-English advisories.
 - Provides actionable recommendations for maintenance teams.
- 
- A decorative wireframe grid pattern in the bottom-right corner of the slide.



THE SOLUTION

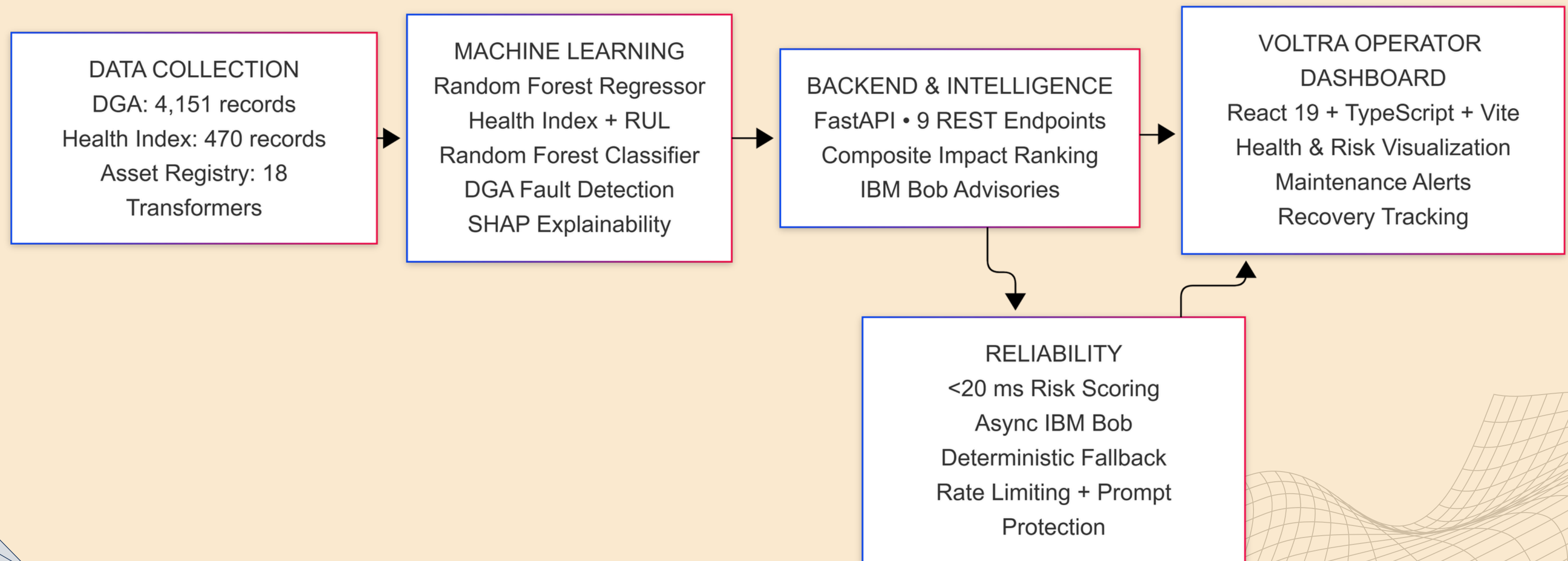
PREDICT. EXPLAIN. ACT.



"We trained on real Kaggle data — not synthetic toys. Both models are validated on held-out test sets with honest metric reporting."

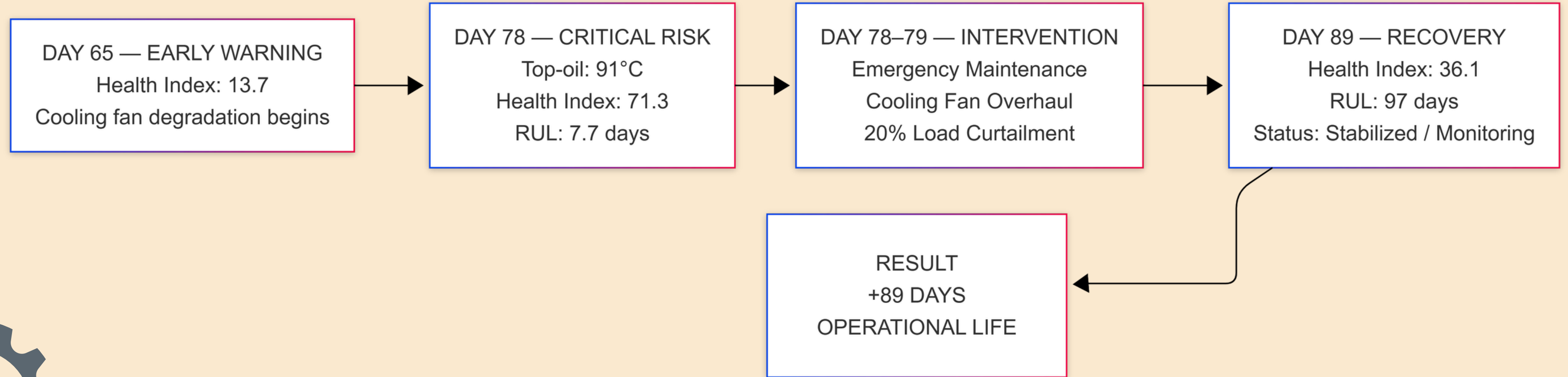
SYSTEM ARCHITECTURE

FROM GRID DATA TO ACTIONABLE INSIGHTS



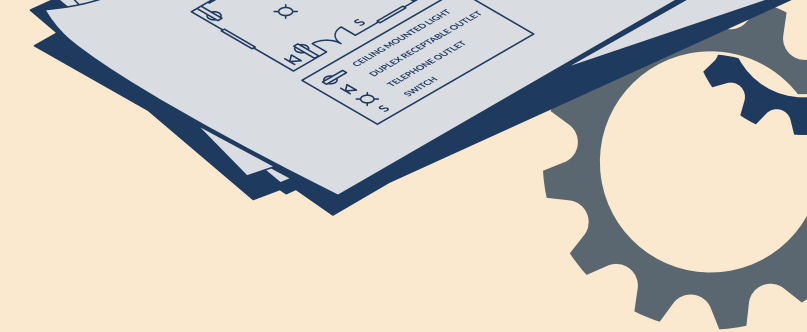
TX-115 INTERVENTION STORY

FROM IMMINENT FAILURE TO RECOVERY



VOLTRA DOESN'T JUST DETECT A FAILURE — IT TRACKS THE RECOVERY AND VERIFIES THAT THE INTERVENTION WORKED.

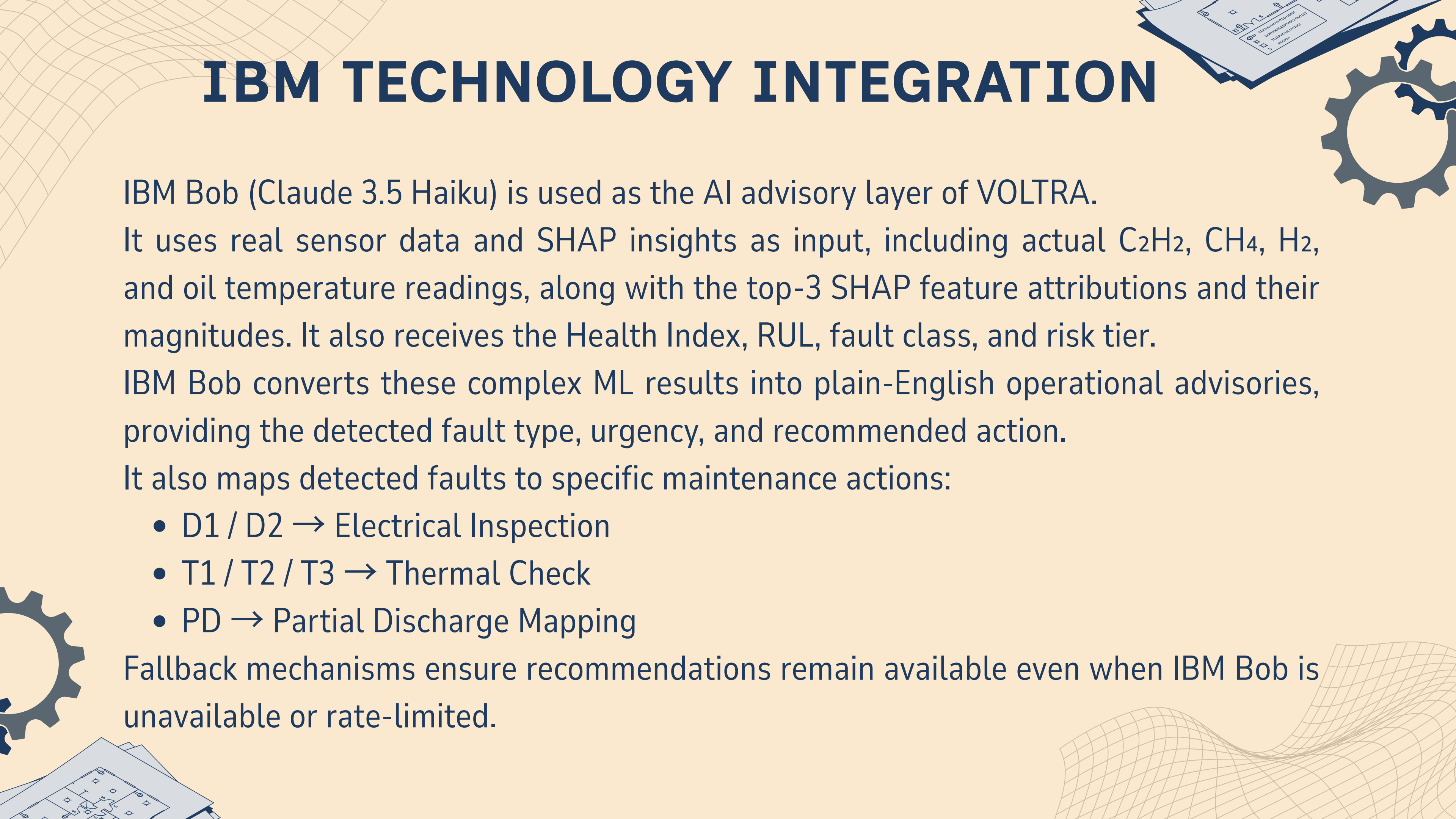
VOLTRA TECHNOLOGY STACK



Layer	Technologies	Purpose
Frontend	React 19, TypeScript, Vite, Tailwind CSS, Recharts	Operator dashboard & visualization
Backend	Python 3.11, FastAPI, Uvicorn	REST APIs & backend services
Machine Learning	Scikit-learn, Random Forest, SHAP, Pandas, NumPy	Health Index, RUL & DGA fault prediction
AI Advisory	IBM Bob, Groq, Gemini	Advisories, forecasting & risk analysis
Database & Auth	MongoDB Atlas, JWT, Google OAuth	Data storage & authentication
External APIs	Open-Meteo	Weather & thermal stress data
DevOps	Docker, GitHub Actions	Deployment & CI/CD
Data Sources	Kaggle, Synthetic Data, Live Weather	Model training & risk analysis



IBM TECHNOLOGY INTEGRATION



IBM Bob (Claude 3.5 Haiku) is used as the AI advisory layer of VOLTRA.

It uses real sensor data and SHAP insights as input, including actual C_2H_2 , CH_4 , H_2 , and oil temperature readings, along with the top-3 SHAP feature attributions and their magnitudes. It also receives the Health Index, RUL, fault class, and risk tier.

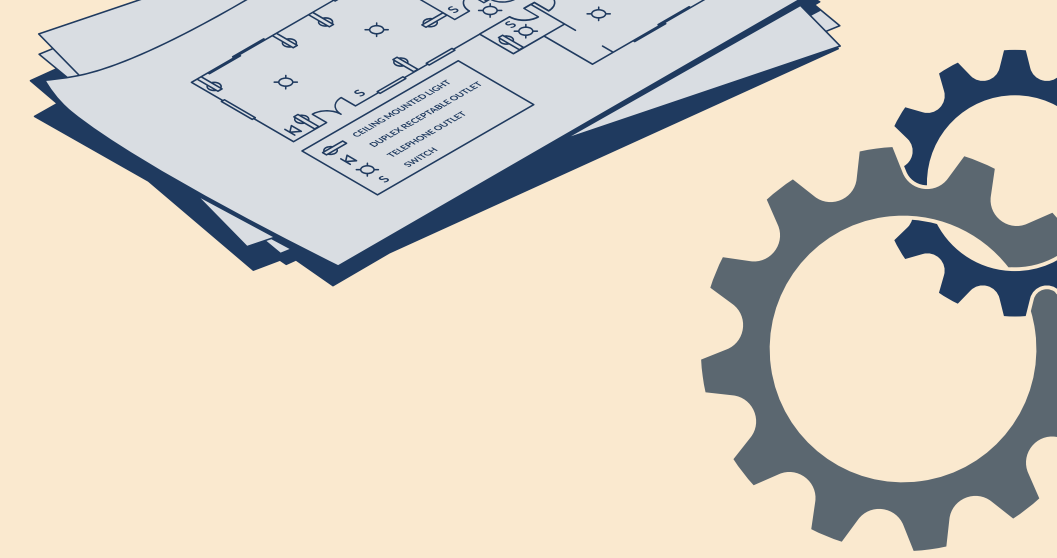
IBM Bob converts these complex ML results into plain-English operational advisories, providing the detected fault type, urgency, and recommended action.

It also maps detected faults to specific maintenance actions:

- D1 / D2 → Electrical Inspection
- T1 / T2 / T3 → Thermal Check
- PD → Partial Discharge Mapping

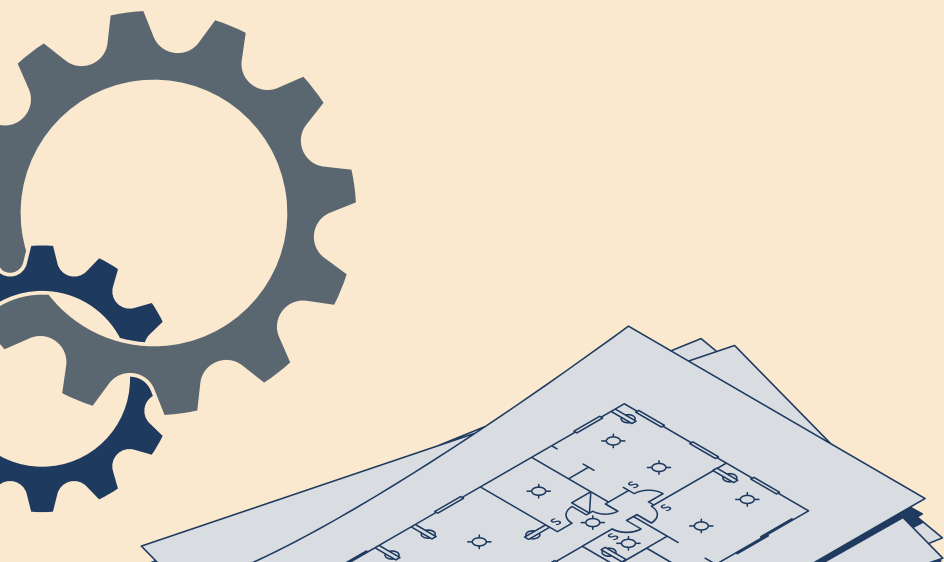
Fallback mechanisms ensure recommendations remain available even when IBM Bob is unavailable or rate-limited.

IMPACT & RESULTS



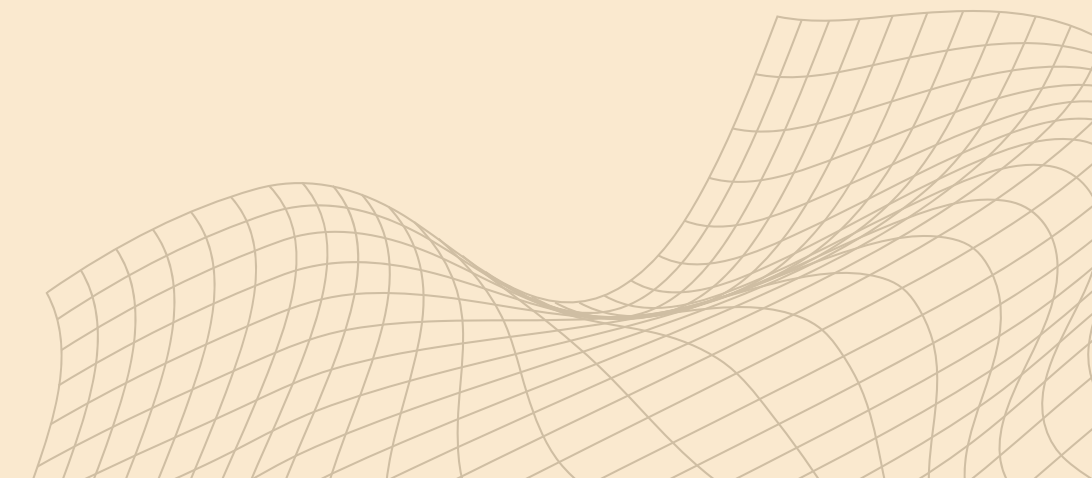
MEASURABLE RESULTS

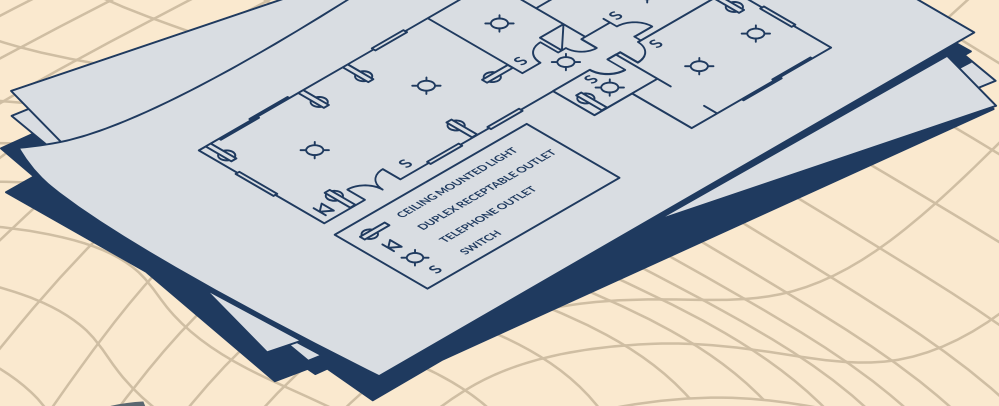
- 90.8% — DGA fault classification accuracy
- R^2 0.72 — Health Index prediction
- <20 ms — Core risk scoring latency
- 38% — Estimated reduction in unplanned downtime
- \$1.8M+ — Potential savings per critical transformer



WHY VOLTRA?

- Predict failures early
- Prioritize the highest-risk assets
- Explain why the risk exists
- Recommend the right action
- Verify recovery after maintenance





TEAM TECHTONICS

Team Member	Role & Contribution
Om Rashiya — Lead	Full-Stack Architecture, FastAPI Backend, React Console, Security Pipeline
Yash Bharvada	Python ML Pipeline, Model Training, IBM Bob & Groq Integration
Nikunj Desai	Frontend Polish, Mobile Responsiveness, UI/UX
Purva Shah	Documentation, Architecture, Presentation, End-to-End Testing

